

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's 41-subject taxonomy includes an Environmental Sciences domain and an Agriculture subject, but both are surface-level and India-scoped by design [Q19, Q38]. The taxonomy lacks any Bangladesh-specific agro-ecological subdomains — no haor/beel wetland category, no char-land farming, no boro/aman rice cycle subject, and no trans-border water management category. Dataset analysis confirms this empirically: across 215 sampled examples, Environmental Sciences is populated almost entirely with geography trivia [DATASET-D6, D9, D16, D18] rather than agro-ecological science, and the few Agriculture-labeled questions concern Indian schemes/institutions [DATASET-D10, D13, D14]. The authors themselves note poor model performance in 'culturally relevant areas' [Q6, Q68] which the deployment population would consider their core expertise. For a HIGH-priority deployment requiring agro-ecological depth across Bangladeshi delta and Indian cross-reference zones, the taxonomy is materially underrepresentative.

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
D6	A geographic region has the following distinct characteristics: 1. Warm and dry climate 2. Mild and wet winter 3. Evergreen oak trees — Mediterranean climate region identification — generic geography
D9	Which state became the first Indian state to launch pension for single women at Rs.1,000/- per month? ... Telangana — Telangana state welfare policy — classified under Geography, reflects generic Indian current affairs
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	राष्ट्रीय मोहरी संशोधन केंद्र कुठे स्थिति आहे... सेवर — Location of National Mustard Research Centre in India — India-specific agricultural institution
D14	ନିମ୍ନଲିଖିତ ଯେଉଁ କମ୍ପାନୀ ଦ୍ରୁତ ଗତିରେ କୃଷି ଉପକରଣ ଉତ୍ପାଦନ କରନ୍ତି?... ଦ କିରଲୋସ୍କାର — Which Indian company produces agricultural equipment rapidly (Kirloskar) — India-specific corporate knowledge
D16	మధ్యప్రదేశ్‌లోని బన్మోర్ పారిశ్రామిక అభివృద్ధి కేంద్రం ఏ జిల్లాలో ఉంది?... మౌరసా — Location of Banmore Industrial Development Centre in Madhya Pradesh — Indian geography
D18	domain: Environmental Sciences, subject: Geography — Mediterranean climate question under Environmental Sciences — geographic reasoning, not agro-ecological depth

Strengths

- MILU's taxonomy does include an Environmental Sciences domain and an Agriculture subject label that appears across multiple Indic languages, providing at least minimal coverage of the relevant domain category [Q38, DATASET-D10, DATASET-D13]

- The 8-domain × 41-subject structure offers broad coverage breadth that enables comparative diagnosis of model capabilities across general knowledge areas, useful as a baseline screening tool [Q1, Q18]

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q18	"This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others." (p.3)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	राष्ट्रीय मोहरी संशोधन केंद्र कुठे स्थिति आहे... सेवर — Location of National Mustard Research Centre in India — India-specific agricultural institution

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include: haor/beel wetland ecology, char-land floodplain agriculture, boro/aman/aus rice cultivation systems, trans-border river hydrology (GBM system), Telangana dry-land cropping (cotton/sorghum/pigeonpea on Deccan red/black soils), Andhra Pradesh coastal aquaculture (L. vannamei/P. monodon under CAA/MPEDA frameworks), and integrated crop-fish farming. None of these appear as named subjects in MILU's 41-subject taxonomy [Q38].

IO-2: Yes — the taxonomy omits all Bangladesh-specific agro-ecological categories. The deployment YAML confirms this is structurally absent (GAP-1 resolution: 'CONFIRMED STRUCTURALLY ABSENT' [WEB-17]), and dataset analysis confirms no haor/beel/char/boro/aman content appears in 215 sampled examples (CRITICAL Concern 1: DATASET-D10, D13, D14, D15).

IO-3: Many MILU subjects are India-centric civic/cultural categories (Indian constitutional law, Indian state polity, medieval Indian history) that introduce construct-irrelevant variance for an environmental-science deployment. Dataset evidence: DATASET-D2 (1991 Indian fiscal crisis), D5 (Indian Election Commissioner removal), D19 (Bahmani Kingdom), D27 (Radcliffe Line), D28 (44th Constitutional Amendment) — none relevant to agricultural/environmental science.

IO-4: The category gap encompasses (a) absent wetland/floodplain ecology subdomains, (b) absent trans-border hydrology subdomain, (c) absent technical agronomy depth (soil science, crop physiology, irrigation) — only policy/institutional trivia appears under 'Agriculture' (CRITICAL Concern 1, MAJOR Concern 5). This constitutes substantial construct underrepresentation for the deployment.

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)

Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D2	১৯৯১ সালে আর্থিক সংকটের কারণ হিসেবে বদেশি মুদ্রার রজিার্ড হ্রাসের তাৎক্ষণিক কারণ কী ছিল? — India's 1991 foreign reserve crisis — Indian economic history, irrelevant to Bangladesh agricultural deployment
D5	নর্বাচন কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নর্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	রাষ্ট্রীয় মোহরী সংশোধন কেন্দ্র কুঠে স্থতি আছে... সেবর — Location of National Mustard Research Centre in India — India-specific agricultural institution
D14	নর্দিল্লিগতি নথ্যর ক্রমে ঠালিও দত্তগতরে কষ্ট্র উপকরণ উত্পাদন করে?... ও করিল্লোয়কার — Which Indian company produces agricultural equipment rapidly (Kirloskar) — India-specific corporate knowledge
D15	রাষ্ট্রীয় খাদ্য সুরক্ষা মসিন ঈব্ ডমল ব্বিাস জেননা ঐ... মর্টি এ সযিও ঠুঁ ব্বাল ব্বনা — National Food Security Mission in Punjabi — same cross-language India-scoped agriculture question
D19	বাহমানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল?... গুলবার্গা — First capital of Bahmani Kingdom — medieval Indian history, no agriculture/environment relevance
D27	রেডক্লিফ লাইন নীচের ঐকী ক্যা দেশ সাথে ভারতের সর্ভদেয়ে অলগ করে ঠে?... পাকিস্তান — Radcliffe Line divides India from Pakistan — Indian partition geography
D28	রাষ্ট্রীয় আপাতকাল ঘোষতি করনে কে লগি 'সশস্ত্র ব্বদ্রোহ' শব্দ সংবধিন মে কব জোড়া গ্যা?... 44বে সংবধিন সংশোধন অধনিয়িম দ্বারা — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)

Information Gaps

- Exact subject-level question counts in Environmental Sciences for Bengali and Telugu test split (Table 9 not directly accessible)
- Whether any Telugu test questions include Telangana/AP sub-regional agro-ecological content (sample of 25 validation examples cannot rule in/out)

Requires Expert Verification

- A Bangladeshi agronomist's assessment of which specific subdomains would be required at minimum for a baseline professional-knowledge benchmark

Remediation

Gap 1: 41-subject taxonomy lacks Bangladesh-specific agro-ecological subdomains (haor wetland ecology, char-land agriculture, boro/aman rice systems, trans-border hydrology)

Recommendation: Collaborate with BAU Mymensingh and BARI/BRRI to define a Bangladesh-extended subject taxonomy with named subdomains for floodplain ecology, wetland farming, and cross-border river systems; construct an extension benchmark covering those subdomains.